

Report on average ticket prices for flights in 2022

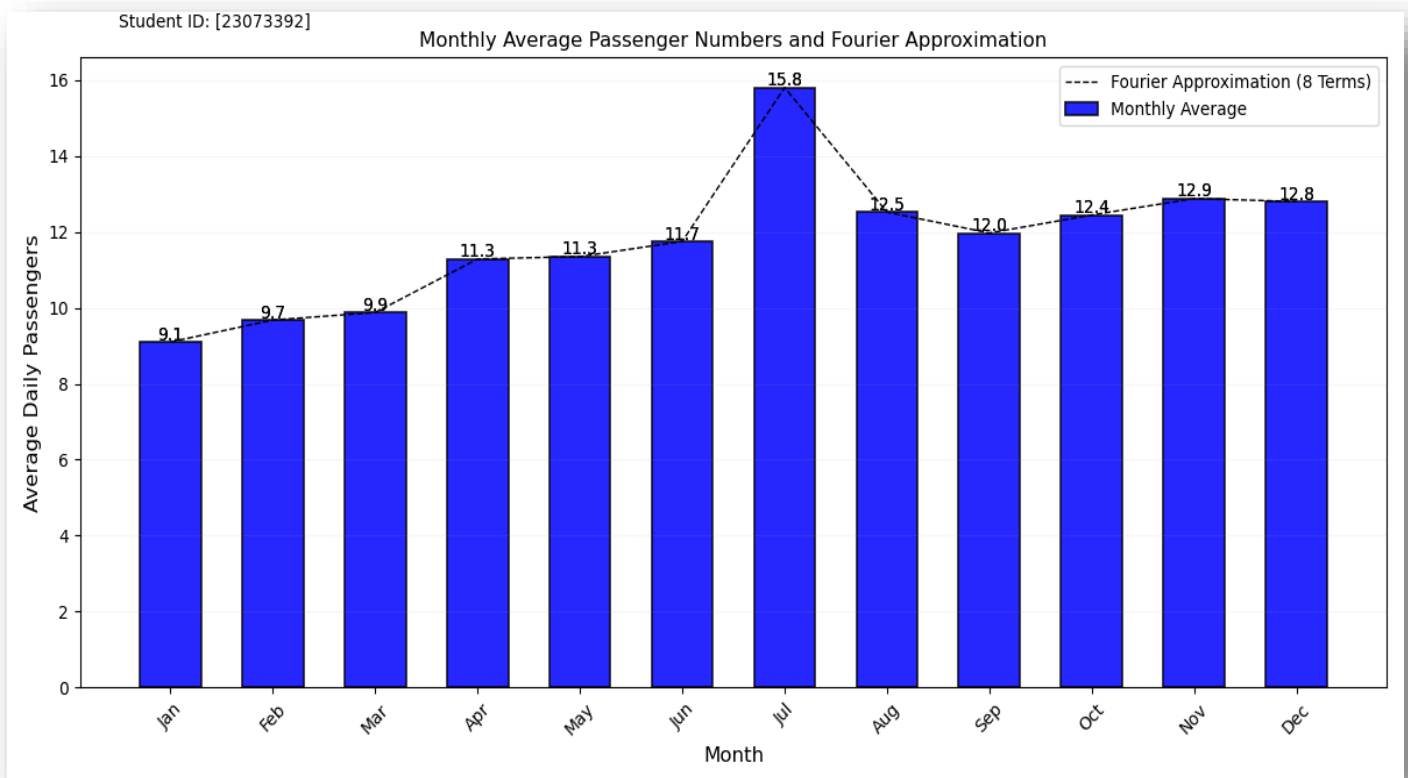
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Introduction :-

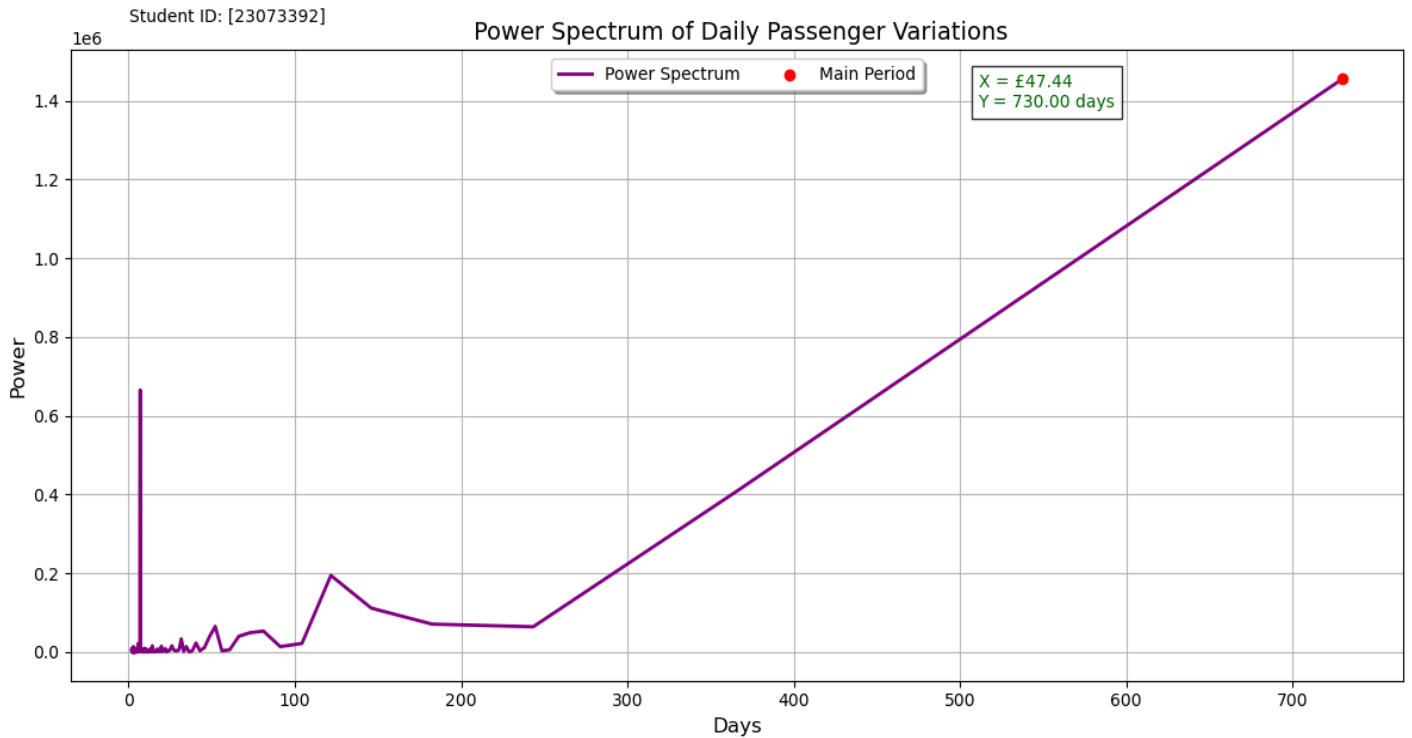
This report mainly uses a dataset related to an airline, which has five variables: day, date, number, price, and revenue. The dataset contains around 730 entries and five columns, and it has no missing values. Data types include integers, floats, and strings. For the visualization bar chart, scatter has been used for more clarity. This report explores average ticket price and monthly average passenger numbers, etc.

Distribution of average daily passengers flown during a year:



In this figure we have taken the X-axis as Month and Y-axis as Average daily passengers, with the use of Fourier series approximation. The Fourier series is used for smooth representations and it highlights the key trend in data and by using eight terms which the approximation simplifies the data into a simpler representation. As we can see the graph is showing a slightly upward trend from Jan till June but in the month of July there is a sudden increase in numbers of daily passengers from 11.7 to 15.8 as well as the important point to be noticed here is after July the number of daily passengers has been decreased to 12.5 and was slightly constant till December. That means July was the peak time where people travelled more through flights. For calculating monthly average I have used mean.

Power Spectra of daily Passenger Variations: -



In this figure we can see the X-axis as Days and Y- axis as Power, where the purple line represents the power spectrum, which shows the distribution of power across different frequencies in the data. The Red Dot indicates the Main Period which is the pattern observed in the passenger variations. In this graph we can see that the power spectrum increases from left to right which means higher frequencies have more power. The calculations which I used in this plotting is of main period which is (Y) that is, $Y = 1/f_{\max}$ where $f_{\max}$ is the maximum power frequency and for calculating the average price ticket for 2022 is $X = \sum 2022 \text{ Price} / N_{2022}$. The Value of $X = £47.44$ and value of $Y = 730$ days.

Conclusion:-

This project successfully analysed the daily passenger number variations using Fourier Transform and Fourier series approximation. The result provide meaning full insights into the patterns and trends in the airline dataset.

